PLANETARY NEBULA

Ant Nebula

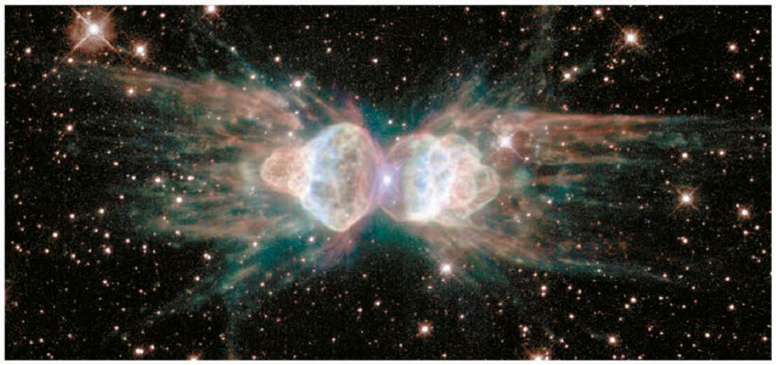
NORMA

CATALOG NUMBER Menzel 3

DISTANCE FROM SUN 4,500 light-years

MAGNITUDE 13.8

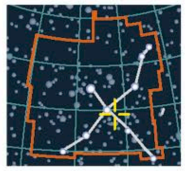
There are two main theories about what has caused the unusual shape of this planetary nebula. Either the central star is a close binary, its interacting gravitational forces shaping the outflowing gas, or it is a single spinning star whose magnetic field is directing the material it has ejected. The expelled stellar material is traveling at around 2.25 million mph (3.6 million kph) and impacting into the surrounding slower-moving medium; the lobes of the nebula stretch to a distance of more than 1.5 light-years. Observations of the Ant Nebula may reveal the future of our own star, because its central star appears to be very similar to the Sun.



HEAD AND THORAX
Even through a small telescope, this planetary nebula resembles the head and thorax of a common garden ant.

PLANETARY NEBULA

Crescent Nebula



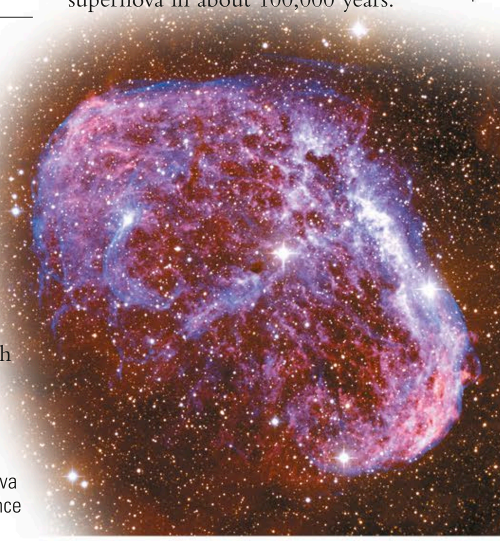
CYGNUS

CATALOG NUMBER NGC 6888

DISTANCE FROM SUN 4,700 light-years

MAGNITUDE 7.44

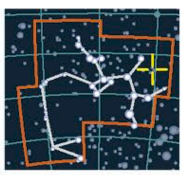
The central star of the Crescent Nebula is a Wolf–Rayet star. Only about 4.5 million years after its formation (one-thousandth the age of the Sun), this massive star expanded to become a red giant and ejected its outer layers at about 22,000 mph (35,000 kph). Two hundred thousand years later, the intense radiation from the exposed, hot inner layer of the star began pushing gas away at speeds in excess of 2.8 million mph (4.5 million kph). This strong stellar wind expelled material equivalent to the Sun's mass every 10,000 years, forming a series of dense, concentric shells that are visible today. Typical of emission nebulae, the radiation from the hot central star excites the stellar material, principally hydrogen, causing it to shine in the red part of the spectrum. It is thought that the nebula's central star will probably explode as a supernova in about 100,000 years.



GASEOUS COCOON
This composite image of the Crescent Nebula shows a compact semicircle of dense material surrounding a presupernova star (center). The Crescent spans a distance of about 3 light-years.

WOLF–RAYET STAR

WR 104

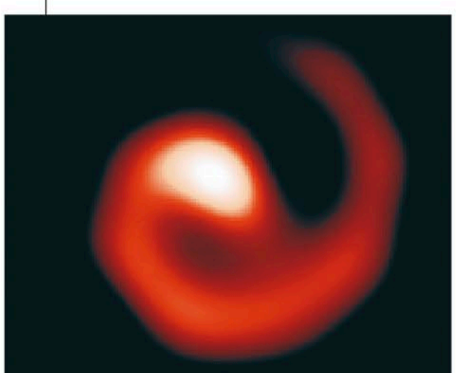


SAGITTARIUS

DISTANCE FROM SUN 4,800 light-years

MAGNITUDE 13.54

SPECTRAL TYPE WCvar+



STELLAR SPIRAL

Like water from a cosmic lawn sprinkler, dust streaming from this rotating star system creates a pinwheel pattern. Since Wolf–Rayet stars are so hot that any dust they emit is usually vaporized, it is surprising that WR 104 has dust streaming away from it in this obvious spiral pattern. One theory is that this is a binary system, with each star emitting a strong stellar wind. Where these winds meet, there is a "shock front" that compresses the outflowing material, creating a denser, slightly cooler environment in which dust can exist. The orbital motion of the two stars then causes the spiral shape.

PLANETARY NEBULA

Eskimo Nebula

GEMINI

CATALOG NUMBER NGC 2392

DISTANCE FROM SUN 5,000 light-years

MAGNITUDE 10.11

The German-born astronomer William Herschel discovered the Eskimo Nebula in 1787, and it has since become a much-loved sight for amateur astronomers. Even through small telescopes, this nebula's form, suggesting a face ringed by a fur parka hood, is clearly visible.

Hubble Space Telescope images reveal a complex structure featuring an inner nebula and an outer halo. The inner nebula consists of material ejected from the central star in two elliptical lobes around 10,000 years before the star was as we now see it. Each lobe is about 1 light-year long and about half a light-year wide and contains filaments of dense matter. Astronomers think that a ring of dense material around the star's equator, ejected during its red-giant phase, helped create the nebula's "face." The surrounding "hood" contains unusual orange filaments, each about 1 light-year long, streaming away from the central star at up to 75,000 mph (120,000 kph). One explanation for these is that they were created when a fast-moving outflow from the central star impacted into slower-moving, previously ejected material.

HOODED NEBULA
In the center of this image, the apparent "face" of the Eskimo consists of one bubble of ejected material lying in front of the other, with the central star visible in the middle.